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Department of Statistics

STAT3003: Survey Methods

Tutorial 9: Applied Problems

1 Random-response model (ref. Textbook p.343-346)

It is rather natural for respondents not to answer or give correct answers to sensitive or embarrassing questions such as “Have you ever taken drugs before?”

To reduce this error due to the respondent’s refusal to answer, we can introduce what is called the random-response model.

Consider a question asking whether the respondent belongs to Group A (e.g. has taken drugs) or Group B (e.g. hasn’t taken drugs). Each respondent is in either Group A or Group B. Our aim is to estimate p , the proportion of the population in Group A, without asking each respondent directly whether he or she belongs to Group A. By not asking directly whether they are in Group A, we hope to reduce or eliminate the stigma/hesitation about answering the question.

1.1 The Procedure

1. Start with a deck of cards, appearing identical except that a proportion θ of them are marked ‘A’ and $1 - \theta$ are marked ‘B’, where θ is a fixed proportion that we chose beforehand.
2. A simple random sample of n people is selected from the population N . Each respondent is asked to randomly draw a card from the deck and say “yes” if the respondent belongs to the Group shown on the card. (Interviewer cannot see what is marked on the card, instead can just record the answer “yes” or “no” from the respondent)
3. The card is replaced before the next respondent draws to keep a fixed θ .

Let n_1 be the number of people in the sample who respond with “yes”. The probability of selecting a card showing Group A and Group B are θ and $1 - \theta$ respectively. Given a card showing Group A, a proportion of p respondents will answer “yes”, and a proportion of $1 - p$ respondents will answer “no”, vice versa in given a card showing Group B. It can be well illustrated using a tree diagram.

The probability of Yes:

$$P(\text{Yes}) = \theta p + (1 - \theta)(1 - p) = p(2\theta - 1) + (1 - \theta)$$

The unbiased estimator of $P(\text{Yes})$ is the sample proportion $P(\widehat{\text{Yes}}) = \frac{n_1}{n}$.

Therefore, $\frac{n_1}{n} = \hat{p}(2\theta - 1) + (1 - \theta)$, or:

$$\hat{p} = \frac{1}{2\theta - 1} \left(\frac{n_1}{n} \right) - \left(\frac{1 - \theta}{2\theta - 1} \right), \quad \theta \neq \frac{1}{2}$$

Variance of $\hat{p}$ can be estimated using methods in SRS variance formula:

$$\widehat{Var}(\hat{p}) = \frac{1}{(2\theta - 1)^2} \widehat{Var} \left(\frac{n_1}{n} \right) = \frac{1}{(2\theta - 1)^2} \left(1 - \frac{n}{N} \right) \left(\frac{1}{n - 1} \right) \left(\frac{n_1}{n} \right) \left(1 - \frac{n_1}{n} \right) \text{ for large } N, n$$

1.2 Illustrative Example

Terry would like to estimate the proportion of CUHK students who have cheated in their mid-term test or final exam during university study before.

- A: You have cheated in mid-term test or final exam during university study before
- B: You have not cheated in mid-term test or final exam during university study before

Terry prepared a deck of cards with three-fourth are marked “A” and one-fourth are marked “B”. A SRS of $n = 500$ people is selected from the whole population of CUHK students. In each separate interview, each sampled respondent is asked to draw a card from the deck and to respond “Yes” if you are in the group shown on the card. (Terry does not know what group the respondent is actually in). The result is $n_1 = 150$ “Yes” response recorded. Help Terry estimates p , the proportion of CUHK students who have cheated in their mid-term test or final exam during university study before.

With $\theta = \frac{3}{4}$,

$$\hat{p} = \frac{1}{2\theta - 1} \left(\frac{n_1}{n} \right) - \left(\frac{1 - \theta}{2\theta - 1} \right) = \frac{1}{\frac{6}{4} - 1} \left(\frac{150}{500} \right) - \left(\frac{1 - \frac{3}{4}}{\frac{6}{4} - 1} \right) = 0.1$$

$$\widehat{Var}(\hat{p}) = \frac{1}{(2\theta - 1)^2} \left(\frac{1}{n} \right) \left(\frac{n_1}{n} \right) \left(1 - \frac{n_1}{n} \right) = \frac{1}{(\frac{6}{4} - 1)^2} \left(\frac{1}{500} \right) \left(\frac{150}{500} \right) \left(1 - \frac{150}{500} \right) = 0.00168$$

1.3 Example of Other formulation of random-response model

- A: Have cheated in mid-term test or final exam during university study before?
- B: Is the last digit of your student ID odd?

Each respondent is asked to flip a fair coin, if it turns a head the respondent answer question A, and vice versa. The interviewer does not see the coin. (Assume the last digit of student ID is random, so the probability of “yes” answer to ‘B’ is $1/2$).

$$P(\text{Yes}) = \frac{p}{2} + \frac{1}{4}$$

$$\hat{p} = 2 \left(\frac{n_1}{n} - \frac{1}{4} \right), \quad \widehat{Var}(\hat{p}) = 4 \left(1 - \frac{n}{N} \right) \left(\frac{1}{n-1} \right) \left(\frac{n_1}{n} \right) \left(1 - \frac{n_1}{n} \right) = \left(\frac{4}{n} \right) \left(\frac{n_1}{n} \right) \left(1 - \frac{n_1}{n} \right) \text{ for large } N, n$$

***** Rule of thumb: Always draw the tree diagram and derive the formulas by yourself!**

2 Post-Stratification

Post-Stratification is stratification after the sample has been selected. Sometimes, post-stratification can be used to adjust the estimation when extra information is given (usually after a SRS).

Idea: Treat the sample as sample from Stratified Sampling setup. Estimation is performed just like usual stratified sampling.

Since $n_1, n_2, \dots, n_L$ are **random** in post-stratification (can change from sample to sample even though n is fixed), additional randomness increases the variance of the estimator.

Using the formula $E(n_i) = nA_i \rightarrow E\left(\frac{1}{n_i}\right) \approx \frac{1}{nA_i} + \frac{1-A_i}{n^2A_i^2}$ by Taylor's expansion, where $A_i = \frac{N_i}{N}, i = 1, \dots, L$.

A good approximation for an unbiased estimator of $V(\bar{y}_{pst})$ is given by

$$\widehat{Var}(\bar{y}_{pst}) \approx \frac{1}{n} \left(1 - \frac{n}{N}\right) \sum_{i=1}^L \frac{N_i}{N} s_i^2 + \frac{1}{n^2} \sum_{i=1}^L \left(1 - \frac{N_i}{N}\right) s_i^2$$

Note: Usually, the second term is simply neglected when n is large.

The increase in variance (the second term) is small when n is large (Order is n^2) (i.e. the second term is small). We also need $n_i s_i^2$ to be relatively large as well, in order for the approximation used to find $\widehat{Var}(\bar{y}_{pst})$ to work. So practically we cannot post-stratify too finely.

3 Adjusting for non-response (ref. Textbook p.351-356)

Nonresponse almost always occurs in sample surveys, which may make our sample become non-representative. Non-respondent is any eligible sampling unit for which data are not obtained because the unit refuses to answer, is "not at home", has a language problem, or does not have enough knowledge. Nonresponse bias comes when non-respondents differ from respondents and make up a large enough proportion of the population that such differences affect survey estimates. Different statistical methods can be used to adjust the weights for nonresponse which can reduce the bias of nonresponse.

The principle of stratification can help adjust the analysis of non-response. And if the stratum sizes N_i are known, post-stratification is used. If they are not known and have to be estimated from sample data, we use weighting-class adjustment.

3.1 Post-stratification

When the response rates are significantly different in different levels of variable (e.g. gender), post-stratification can be used to reflect the “true” population.

When the stratum sizes (N_i 's) are known, the adjusted estimate of population mean can be calculated as follow,

$$\bar{y}_{pst} = \frac{N_1}{N} \bar{y}_1 + \frac{N_2}{N} \bar{y}_2 + \cdots + \frac{N_L}{N} \bar{y}_L$$

What if we don't know N_i 's'?

Weight-class adjustment: Estimate $\hat{N}_i = N \times \frac{n_i}{n}$ (include the non-respondents in n_i).

4 Subpopulation (ref. Textbook p.339-342)

Sometimes, the sampling frame may subject to the problem of coverage, for example, we may wish to sample statistics major undergraduates but the best frame available may be a list of containing all statistics department students. There are some ineligible units. In this case, we are actually estimating parameter(s) of a subpopulation of the population represented in the frame. However, we only know whether a respondent/observation is a member of the subpopulation after it is sampled.

Notation:

N	population size
N_1	subpopulation size
n	sample size of the simple random sample drawn from the population
n_1	number of sampled elements in the subpopulation
y_{1j}	j th sampled element in the subpopulation

		μ_1	τ_1
known N_1	Estimator	$\bar{y}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} y_{1j}$	$\frac{N_1}{n_1} \sum_{j=1}^{n_1} y_{1j}$
	Estimated variance	$\widehat{Var}(\bar{y}_1) = \frac{N^2}{N_1^2} \left(1 - \frac{n}{N}\right) \frac{1}{n} \frac{n_1 - 1}{n - 1} s_1^2$	$N_1^2 \times \text{LHS}$
unknown N_1	Estimator	$\bar{y}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} y_{1j}$	$\frac{N}{n} \sum_{j=1}^{n_1} y_{1j}$
	Estimated variance	$\widehat{Var}(\bar{y}_1) = \left(1 - \frac{n}{N}\right) \left(\frac{n}{n-1}\right) \frac{n_1 - 1}{n_1} \frac{s_1^2}{n_1}$ $\approx \left(1 - \frac{n}{N}\right) \frac{s_1^2}{n_1}$ n and n_1 are large	$N^2 \left(1 - \frac{n}{N}\right) \left(\frac{s_n^2}{n}\right)$

where $s_1^2 = \frac{1}{n_1 - 1} \sum_{j=1}^{n_1} (y_{1j} - \bar{y}_1)^2$ and $s_n^2 = \frac{1}{n - 1} \sum_{i=1}^n (y_i - \bar{y})^2 = \frac{1}{n - 1} (\sum_{i=1}^n y_i^2 - n(\bar{y})^2)$.

$\frac{n_1}{N_1}$ can be estimated by $\frac{n}{N}$ if N_1 is unknown. s_n^2 is the sample variance calculated from an adjusted sample consisting of replacing all the observation not from the subpopulation of interest with zeros. Then the sample variance is calculated from the adjusted sample with size n .

Exercises

1. (Adapted from Exercise 5.30 in Scheaffer et al. (2012)) In the late 1990s a sample survey of people emigrating from Cuba to Florida was conducted to obtain their opinions on conditions in their homeland. Among the 1001 sampled individuals, 87% were white and 13% were African American. It is known from census data on Cuba that the country has approximately 60% whites and 40% African Americans. One of the questions asked was “In your opinion, has the Revolution attained accomplishments in education?”. Table 1 shows the results of the survey.

	Yes	No	Total
White	566	304	870
Black	102	29	131
Total	668	333	1001

Table 1: Cuba Survey Data

- Assuming that the 1001 responses came from a SRS of emigrants, estimate the true proportion of emigrants who would answer “Yes” to this question and provide a 95% confidence interval.
- Construct a post-stratification estimate of the true proportion of “Yes” answers among emigrants from Cuba, with an appropriate standard error. Comment on whether or not post-stratification has worked in this situation.

- (Adapted from Exercise 11.13 in Scheaffer et al. (2012)) A public health official wants to estimate the proportion of dog owners in a city who have had their dogs vaccinated against rabies. She knows that a dog owner often gives incorrect information about

rabies vaccinations out of fear that something might happen to their dog if it has not had the vaccination. Thus, the official decides to use a random-response technique. She has a stack of cards with 0.8 of the cards marked A for the group having the vaccinations and 0.2 marked B for the group not having the vaccinations. A SRS of 200 dog owners is selected. Each sampled owner is interviewed and asked to draw a card and to respond with “Yes” if the letter on the card agrees with the group he is in. The official obtained 145 “Yes” responses. Estimate the proportion of dog owners who have had their dogs vaccinated and provide a 95% confidence interval. Assume that the number of dog owners in the city is very large.

3. (Adapted from Exercise 11.11 in Scheaffer et al. (2012)) A survey of statistics departments collected data on enrollments in large lecture sections of introductory statistics courses, as well as the number of discussion sections, for the 2016 autumn semester. The data are grouped into five strata after the selection of a SRS of 70 statistics departments. Strata 1 through 4 are PhD-granting departments in universities with size boundaries of 15,000, 25,000, 35,000 students. Stratum 5 includes all departments that do not grant a PhD degree. In addition, suppose the sizes of the five strata are not known. The sample data on enrollments are provided in Table 2. The actual sample sizes in the five post-strata were 9, 14, 7, 10 and 5 for an overall sample size of 45. Using an adjustment for non-response, estimate the total number of students taking large lecture courses in elementary statistics in the autumn of 2016, and find a standard error for your estimate.

By treating these data as from a stratified random sample and using the usual stratified random sampling formulae (with strata sizes known), it was found that the estimate of the total number of students enrolled in these large-section courses (in statistics departments in the autumn of 2016 is $70 \times 656.774 = 45,974$. The standard error was approximately 4500. Compare your non-response-adjusted estimate with this.

Stratum 1 # enrolled	Stratum 2 # enrolled	Stratum 3 # enrolled	Stratum 4 # enrolled	Stratum 5 # enrolled
494	1015	784	1357	1557
36	700	1101	1433	88
153	1391	285	1680	500
277	101	255	934	390
480	1100	755	615	
500	731		834	
	689		506	
	227			
	151			
	375			

Table 2: Statistics Departments Survey

4. (Adapted from Exercises 11.6 and 11.7 in Scheaffer et al. (2012)) A study is conducted to estimate the average number of miles from home to place of employment for household heads living in a certain suburban area. A SRS of 30 people is selected from the 493 heads of households in the area. While conducting interviews, the experimenter finds some household heads are not appropriate for the study because they either are retired or do not go to a place of employment for various reasons. Of the 30 sampled household heads, 24 are appropriate for the study, and the data on miles to place of employment are given in Table 3.

8.5	10.2	25.1	5.0	6.3	7.9	15.8	2.1
9.2	4.2	8.3	4.2	6.7	10.1	15.6	22.1
10.0	6.1	7.9	1.5	8.0	11.0	20.2	9.3

Table 3: Household Heads Survey Data

- Estimate the average distance between home and place of employment for household heads who commute to a place of employment. Provide a 95% confidence interval.
- Estimate the total travel distance between home and place of employment for all household heads in the suburban area. Provide a 95% confidence interval.

THE END